

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – IX
PAPER –2
TEST DATE: 17-08-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, B, C, D
 Sol. For the stars

$$\frac{GM^2}{(4R)^2} = M(2R)\omega^2$$

$$\omega = \sqrt{\frac{GM}{32R^3}} \quad \dots(i)$$

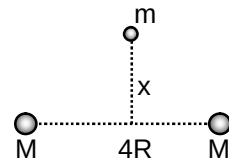
The asteroid should be equidistant from both the stars.

$$\frac{2GMmX}{[(X^2 + (2R)^2)^{3/2}]} = mX\omega^2 \quad \dots(ii)$$

From (i) and (ii)

$$X = 2\sqrt{3}R$$

$$\text{Total energy} = -\frac{2GMm}{4R} + \frac{1}{2}m\left(2\sqrt{3}R\sqrt{\frac{GM}{32R^3}}\right)^2 = -\frac{5GMm}{16R}$$



2. A, B, C
 Sol. $x = t^3 - 6t^2 - 96t + 37$... (i)
 $v = 3t^2 - 12t - 96$... (ii)
 $a = 6t - 12$... (iii)

at $t = 0$, $v = -96$ m/s

when $v = 0$, $t = 8$ sec

and when v is maximum $t = 2$ sec.

3. A, B, C

Sol. Percentage error in a , b and M are $\frac{\Delta a}{a} \times 100 = \frac{0.001}{4} \times 100 = 0.025\%$

$$\frac{\Delta b}{b} \times 100 = \frac{0.001}{2} \times 100 = 0.05\%$$

$$\frac{\Delta M}{M} \times 100 = \frac{0.0003}{0.3} \times 100 = 0.1\%$$

Moment of Inertia

$$I = \frac{M}{2}(a^2 + b^2)$$

$$\frac{\Delta I}{I} = \frac{\Delta M}{M} + \frac{2a \cdot \Delta a + 2b \cdot \Delta b}{a^2 + b^2}$$

$$\frac{\Delta I}{I} = 0.001 + \frac{2(4)(0.001) + 2(2)(0.001)}{20}$$

$$\frac{\Delta I}{I} = 0.001 + \frac{0.008 + 0.004}{20} = 0.0016$$

$$\frac{\Delta I}{I} \times 100 = 0.16\%$$

4. B, D

Sol. As we know that,

$$\frac{hC}{\lambda} = \phi + K \Rightarrow \frac{12400}{\lambda} = \phi + v_s, \text{ where } \phi \text{ is in electron Volt and } \lambda \text{ is in } \text{\AA}$$

When light of wavelength $\lambda = 6000 \text{ \AA}$ is used

$$\frac{12400}{6000} = \phi + 0.3 \Rightarrow \phi = 1.77 \approx 1.78 \text{ eV (approximately)}$$

When light with wavelength $\lambda = 4000 \text{ \AA}$ is used

$$\frac{12400}{4000} = \phi + eV$$

$$\Rightarrow eV = (3.1 - 1.78) \text{ e volt} = 1.32 \text{ eV}$$

$$\Rightarrow V = 1.32 \text{ Volt}$$

5. B, C

Sol. Heat gain by ice to melt = $mL_f = 10 \times 80 = 800 \text{ cal}$

Heat gain by 10 g water to raise its temperature from 0°C to $100^\circ\text{C} = 10 \times 1 \times 100 = 1000 \text{ cal}$

Total heat gain = 1800 cal

Mass of steam converted into water

$$\therefore 1800 = m \times 540 \Rightarrow m = 3.33 \text{ g}$$

$\therefore$ equilibrium temperature 100°C

Amount of water = $10 + 3.33 = 13.33 \text{ g}$

Amount of steam = $5 - 3.33 = 1.67 \text{ g}$

6. A, B, C, D

Sol. (A) $i_{\text{rms}} = \frac{80}{4} = 20 \text{ amp}$, $i_{\text{peak}} = 20\sqrt{2} \text{ amp}$

$$(B) \varepsilon_{\text{rms}} = \sqrt{(40)^2 + (40 - 10)^2} = 100\text{V}, \varepsilon_0 = 100\sqrt{2}\text{V}$$

$$(C) \omega L i_{\text{rms}} = 90 \Rightarrow L = \frac{90}{10 \times 90\pi} = \frac{1}{20\pi} \text{ H}$$

(D) For maximum heat, current through the circuit will be maximum $\omega L = \frac{1}{\omega C}$

$$C = \frac{1}{405\pi} \text{ F}$$

7.

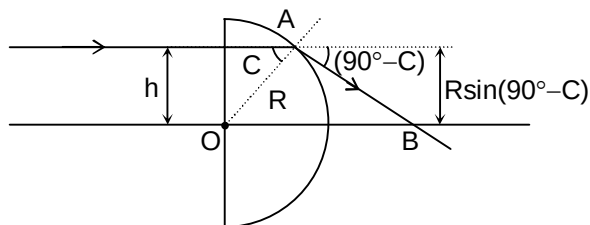
C

Sol. $\frac{5}{3} \sin C = 1 \sin 90^\circ$

$$\sin C = 3/5 \quad (C \rightarrow \text{critical angle})$$

$$C = 37^\circ$$

$$h = R \sin 37^\circ = \frac{3}{5} R = \frac{3}{5} (30) = 18 \text{ cm}$$



8.

C

Sol. when $\mu = \sqrt{2}$

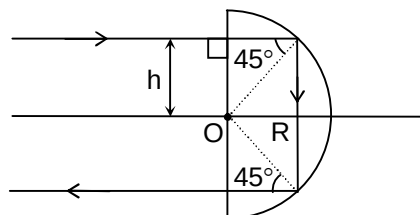
$$\sqrt{2} \sin C = 1 \quad (C \rightarrow \text{critical angle})$$

$$C = 45^\circ$$

Total distance travel in semi-cylinder

$$= \left(\frac{R}{\sqrt{2}} \right) 4 = 2\sqrt{2}R = (2\sqrt{2})(30) = 60\sqrt{2} \text{ cm}$$

$$\text{Time} = \frac{60\sqrt{2}}{C_m} \times 10^{-2} = \frac{60\sqrt{2} \times 10^{-2}}{\left(\frac{3 \times 10^8}{\sqrt{2}} \right)} = \frac{120 \times 10^{-2}}{3 \times 10^8} = 4 \times 10^{-9} \text{ s} \quad (C_m \rightarrow \text{speed of light in the medium})$$



9.

D

10.

D

Sol. (for Q. 9-10):

$$dm = \mu dy$$

$$m = \int_0^y \mu_0 \left(\frac{y}{\ell} \right)^3 dy = \left(\frac{\mu_0}{\ell^3} \right) \left(\frac{y^4}{4} \right)$$

Velocity of wave

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{gy}{4}}$$

$$\frac{dy}{dt} = \frac{\sqrt{gy}}{2}$$

$$\int_0^\ell \frac{dy}{\sqrt{y}} = \frac{\sqrt{g}}{2} \int_0^t dt$$

$$t = \left(4 \sqrt{\frac{\ell}{g}} \right)$$

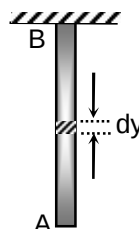
$$v = \frac{\sqrt{gy}}{2}$$

$$v^2 = \frac{gy}{4}$$

$$2v \frac{dv}{dy} = \frac{g}{4}$$

$$v \frac{dv}{dy} = \left(\frac{g}{8} \right)$$

$$a = g/8$$



Section – B

11. 5

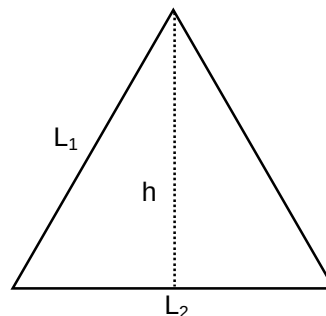
Sol.
$$h^2 = L_1^2 - \frac{L_2^2}{4}$$

 Differentiating and using $dL = L \alpha \Delta t$

$$0 = 2L_1\alpha_1 L_1 \Delta t - \frac{1}{4} 2L_2\alpha_2 L_2 \Delta t$$

$$L_1 L_1 \alpha_1 \Delta t = \frac{1}{4} L_2 L_2 \alpha_2 \Delta t$$

$$\Rightarrow \frac{L_2}{L_1} = 2 \sqrt{\frac{\alpha_1}{\alpha_2}} = 2 \sqrt{\frac{25 \times 10^{-6}}{4 \times 10^{-6}}} = 5$$



12. 10

Sol. For lens, $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

 Here, $R_2 = +60$ cm and $R_1 = 20$ cm

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{60} \right)$$

$$\Rightarrow \frac{1}{60} = \frac{1}{v} + \frac{1}{80}$$

$$v = 240$$

$$\text{Shift due to slab, } s = 2 \left[1 - \frac{1}{1.5} \right] = 1 \text{ cm}$$

For final image to form at O, ray should retrace its path after reflection from mirror. So, I_2 should be at centre of curvature of mirror i.e.

$$7 + 3 + x = 20$$

$$x = 10 \text{ cm}$$

13. 2100

 Sol. $v_w \rightarrow$ velocity of wind

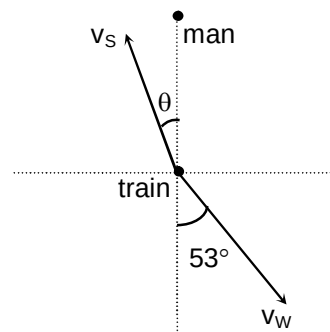
 $v_s \rightarrow$ velocity of sound

$$v_s \sin \theta = v_w \sin 53^\circ$$

$$\sin \theta = \frac{v_w}{v_s} \left(\frac{4}{5} \right) = \frac{100}{(400/3)} \times \frac{4}{5} = \frac{3}{5}$$

$$\theta = 37^\circ$$

$$f' = f \left(\frac{v_s \cos \theta - v_w \cos 53^\circ}{v_s \cos \theta - v_w \cos 53^\circ - v_T} \right) = 300 \times 7 = 2100 \text{ Hz}$$



Section – C

14. 06.00

15. 15.00

Sol. (for Q. 14-15):

By the conservation of linear momentum between m_1 and m_2 and by using coefficient of restitution velocity m_2 just after collision $\frac{3v_0}{4}$.

After the collision, the maximum extension in the spring occurs when angular velocity of m_2 and m_3 about O become same. Now by conservation of angular momentum about 'O' and energy conservation

$$\text{Velocity of } m_2 = 2R\omega = \frac{3}{5}v_0$$

$$\text{Velocity of } m_3 = R\omega = \frac{3}{10}v_0 \text{ and } \Delta x \text{ maximum} = \frac{3}{4}v_0\sqrt{\frac{m}{5k}}$$

16. 10.00

17. 06.28

Sol. (for Q. 16-17):

$$i\left(\frac{8y_0}{3}\right)B = mg$$

$$y_0 = \frac{3mg}{8iB}$$

$$y_0 = 10$$

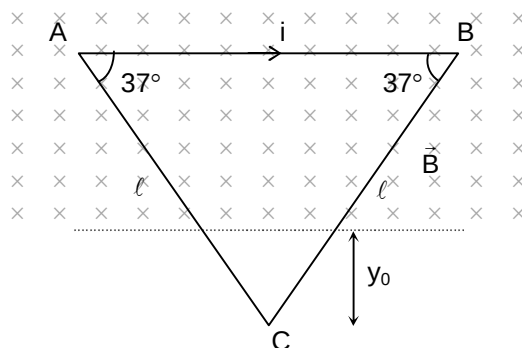
If loop is displaced by small y downward. Then,

$$ma = mg - iB\left[\frac{8}{3}(y + y_0)\right]$$

$$a = -\left(\frac{8iB}{3m}\right)y$$

$$\omega = \sqrt{\frac{8iB}{3m}}$$

$$T = \frac{2\pi}{\omega} = 2\pi$$



18. 09.00

19. 15.00

Sol. (for Q. 18-19):

Difference of potential energy of two configuration when point charge is at rest at a distance d to when it passes through centre.

The energy of oscillation system

$$E = \frac{2KQ^2}{3\sqrt{R^2 + d^2}} - \frac{2KQ^2}{3R} = \frac{Q^2 d^2}{12\pi\epsilon_0 R^3}$$

$$|m - n| = 12 - 3 = 9$$

Force between the ring and the charge particle

$$F = -\left(\frac{2KQ^2}{3R^3}\right)d$$

$$M_{eq} = \frac{2}{3}M$$

$$\text{Hence time period of oscillation is } T = \sqrt{\frac{16\pi^3\epsilon_0 MR^3}{Q^2}}$$

Chemistry

PART – II

Section – A

20. A, B, C

Sol. E_{cell}° , E_{cell} and normality are intensive property, where as entropy is extensive property.

21. B, C

Sol. $[\pi_{\text{obs}}]_{0.4 \text{ M NaCl}} = i \times C \times R \times T$
 $= i \times 0.4 \times R \times T$
 $= [1 + (n - 1)\alpha] \times 0.4 \times R \times T$
 $= [1 + 0.8] \times 0.4 \times R \times T$
 $= 1.8 \times 0.4 \times R \times T$
 $= 0.72 RT$
 $[\pi_{\text{obs}}]_{0.3 \text{ M Na}_3\text{PO}_4} = [1 + (4 - 1) \times 0.9] \times 0.3 \times R \times T$
 $= 1.11 RT$
 $[\pi_{\text{obs}}]_{0.7 \text{ M Glucose}} = 0.7 \times R \times T$
 $= 0.7 RT$
 $[\pi_{\text{obs}}]_{1 \text{ M } \text{HOOC}-\text{C}(=\text{O})-\text{OH}} = [1 + (2 - 1)0.3] \times R \times T$
 $= 1.3 \times R \times T$
 $[\pi_{\text{obs}}]_{0.5 \text{ M MgCl}_2} = 0.5 \times R \times T [1 + (2) \times 0.2]$
 $= R \times T \times 0.5 [1.4]$
 $= 0.7 RT$

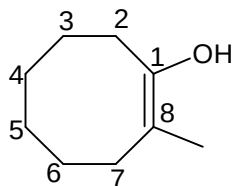
Here only option (B) and (C) have greater osmotic pressure then given solution.

22. A, B, C, D

Sol. In isothermal ideal gas compression as volume decrease. Hence, W is +ve, ΔH is zero, ΔS_{gas} is – ve and ΔE is zero

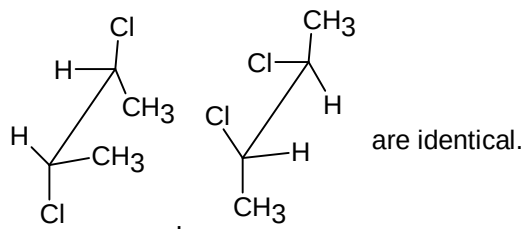
23. B, C

Sol. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$ will show geometrical isomerism.



will show geometrical isomerism

24. A
Sol.

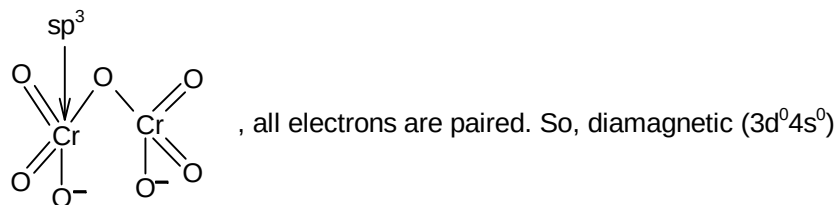


25. A, C, D

- Sol. (A) $\ell = 0$, then $m \neq 1$. Hence, (A) is incorrect.
 (B) It is the correct set.
 (C) If $n = 1$, then $\ell = -1$. Hence, (C) is not correct.
 (D) If $n = 3$, then $m \neq 3$. Hence, (D) is also incorrect.

26. D

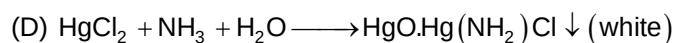
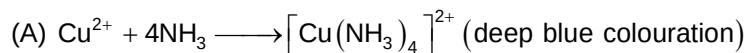
Sol.



$\text{Cr}_2\text{O}_7^{2-}$ acts as strong oxidizing agent in acidic medium.

27. D

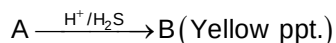
Sol. (E) = NH_3



28. B

29. A

Sol. (for Q. 28 to 29):

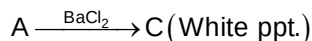


Only CdS and As_2S_3 are yellow

CdS insoluble in Yellow Ammonium Sulphide

As_2S_3 soluble in Yellow Ammonium Sulphide

So, B is CdS



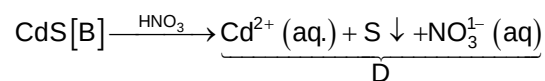
BaSO_4 are white ppt.

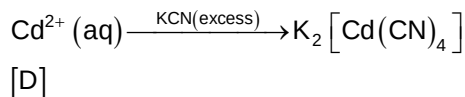
BaSO_4 Insoluble in mineral acids

BaSO_3 soluble in mineral acids

So, C is BaSO_4

$\therefore$ A is CdSO_4





Section – B

30. 2

Sol. $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{PtCl}_3\text{NH}_3]$

31. 6

Sol. Number of stereo isomers for the complex is 6.

32. 9

Sol. $3x_2(\text{g}) \rightleftharpoons x_6(\text{g}), K_p = 1.6 \text{ atm}^{-2}$

$$2a - x - z \quad \frac{x}{3}$$

$$x_2(\text{g}) + y(\text{g}) \rightleftharpoons x_2y(\text{g}), K_p = W \text{ atm}^{-1}$$

$$2a - x - z \quad a - z \quad z$$

$$\frac{x}{3} = 0.2 \Rightarrow x = 0.6$$

$$1.6 = \frac{P_{x_6}}{[P_{x_2}]^3} \Rightarrow [P_{x_2}]^3 = \frac{0.2}{1.6}$$

$$P_{x_2} = \frac{1}{2} = 0.5 \text{ atm}$$

$$\text{Now, } 0.5 + 0.2 + a - z + z = 1.4$$

$$a = 0.7$$

$$2a - x - z = 0.5$$

$$\Rightarrow 2 \times 0.7 - 0.6 - 0.5 = z$$

$$z = 0.3$$

$$K_p = W = \frac{0.3}{0.5 \times [0.7 - 0.3]} = 1.5$$

$$\text{Hence, } 6W = 6 \times 1.5 = 9$$

Section – C

33. 166.67

Sol. 1st order kinetic

$$t = \frac{2.303}{k} \log \frac{a_0}{a_t}$$

$$\frac{0.6932}{50} = \frac{2.303}{t} \log \frac{100}{10}$$

34. 00.07

Sol. $\text{A}(\text{g}) \longrightarrow 3\text{B}(\text{g}) + 2\text{C}(\text{g})$

$$t = 0 \quad \left[\frac{2}{12.5} \right] \quad 0 \quad 0$$

$$t = t \quad \frac{2-x}{V} \quad \frac{3x}{V} \quad \frac{2x}{V}$$

$$\begin{aligned}\text{Now, } t &= \frac{2.303}{K} \log \frac{a_0}{a_t} \\ \Rightarrow 100 &= \frac{2.303}{1.386 \times 10^{-2}} \log \frac{2/12.5}{a_t} \\ \Rightarrow at &= 0.04\end{aligned}$$

$$\text{Now, } \frac{2-x}{V} = 0.04 \quad \dots (1)$$

And as the P, T constant

$$\frac{2}{12.5} = \frac{2+4x}{V} \quad \dots (2)$$

On solving Eq. (1) and (2), we get

$$[B_t] = 0.07 \text{ M}$$

35. 17.72

Sol. Meq. of NH_3 = Meq. of HCl – Meq. of NaOH

$$\frac{W \times 1000}{17} = 5 - 2.32 = 2.68$$

$$W = \frac{17 \times 2.68}{1000} = 0.4556$$

$$\begin{aligned}\%N &= \frac{0.4556}{0.257} \times 100 \\ &= 17.72\%\end{aligned}$$

36. 34.04

$$\text{Sol. } \%Br = \frac{80}{188} \times \frac{0.12}{0.15} \times 100 = 34.04\%$$

37. 03.00

$$\text{Sol. } E = 0.61 = 0.76 - \frac{0.06}{2} \log \frac{[Zn^{2+}]}{[H^+]^2}$$

$$\Rightarrow \log \frac{(0.1)}{[H^+]^2} = \frac{0.76 - 0.61}{0.03}$$

$$\Rightarrow \text{pH} = 3$$

38. 117.73

Sol. $\Delta G = -nFE_{\text{cell}}$

$$\begin{aligned}|\Delta G| &= \frac{2 \times 0.61 \times 96500}{1000} \text{ kJ} \\ &= 117.73 \text{ kJ}\end{aligned}$$

Mathematics

PART – III

Section – A

39. A, C

$$\text{Sol. } \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \frac{k-1}{k!} \right) = \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \left\{ \frac{1}{(k-1)!} - \frac{1}{k!} \right\} \right)$$

$$L = \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \left(1 - \frac{1}{i!} \right) \right); L = \lim_{n \rightarrow \infty} \left(2 + \sum_{i=2}^n \frac{1}{i!} \right)$$

$$\therefore L = e$$

40. B, D

Sol. Let $g(0) = c$ then put $x = x$ and $y = 0$ in functional equation $f(x + g(y)) = 2x + y + 5$

$$\Rightarrow f(x + c) = 2x + 5 \Rightarrow f(x) = 2x + 5 - 2c$$

$$\Rightarrow f(x + g(y)) = 2x + 2(g(y)) - 2c + 5 \text{ compare with give equation}$$

$$\Rightarrow g(y) = \frac{y}{2} + c \Rightarrow g(x + f(y)) = \frac{x + f(y)}{2} + c \Rightarrow g(x + f(y)) = \frac{x + 2y - 2c + 5}{2} + c = \frac{x + 2y + 5}{2}$$

41. C

$$\text{Sol. } {}^m C_1 \cdot {}^n C_m - {}^m C_2 \cdot {}^{2n} C_m + {}^m C_3 \cdot {}^{3n} C_m \dots (-1)^{m-1} \cdot {}^m C_m \cdot {}^{mn} C_n$$

$\Rightarrow$ Coefficient of x^m in expression

$${}^m C_1 \cdot (1+x)^n - {}^m C_2 \cdot (1+x)^{2n} + {}^m C_3 \cdot (1+x)^{3n} \dots {}^m C_m \cdot ((1+x)^{mn}) \cdot (-1)^{m-1}$$

$$\Rightarrow {}^m C_0 - [{}^m C_0 - {}^m C_1 \cdot (1+x)^n + {}^m C_2 \cdot (1+x)^{2n} - {}^m C_3 \cdot (1+x)^{3n} \dots {}^m C_m \cdot (-1)^{mn} \cdot (1+x)^{mn}]$$

$$\Rightarrow \text{Coefficient of } x^m \text{ in } {}^m C_0 - (1 - (1+x)^n)^m \Rightarrow \text{Coefficient of } x^m \text{ in } (1 - (1 - (1+x)^n)^m)$$

42. B, D

Sol. If there is $2k$ occurrence of a then this can be done by ${}^n C_{2k}$ ways and remaining positions can be

$$\text{filled in } 2^{n-2k} \text{ ways. So total number of ways are } \sum_{k=0}^n {}^n C_{2k} 2^{n-2k} = \frac{1}{2}(3^n + 1)$$

43. B, C

$$\text{Sol. Let } f(x) = \frac{1}{1+x}, \text{ then } \frac{1}{n} \left(\frac{1}{1+\frac{1}{n}} \right) < \int_0^{\frac{1}{n}} f(x) dx < \frac{1}{n}, \frac{1}{1+n} < \ln \left(1 + \frac{1}{n} \right) < \frac{1}{n}$$

$$\text{Now, } a_n - a_{n+1} = -\log n - \frac{1}{n+1} + \log(n+1) = \log \left(1 + \frac{1}{n} \right) - \frac{1}{n+1} > 0$$

$$b_{n+1} - b_n = a_{n+1} - a_n - \frac{1}{n+1} + \frac{1}{n} = \frac{1}{n+1} - \log \left(1 + \frac{1}{n} \right) - \frac{1}{n+1} + \frac{1}{n} = -\log \left(1 + \frac{1}{n} \right) + \frac{1}{n} > 0$$

44. A, C, D

Sol. Let coordinate of A(a, 9 - a), then the distance of the centre of circle

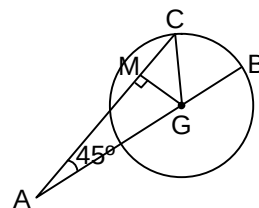
$$x^2 + y^2 - 4x - 4y - \frac{1}{2} = 0 \text{ from AC will be}$$

$$= |AG| \sin 45^\circ$$

$$= \sqrt{(a-2)^2 + (7-a)^2} \cdot \frac{1}{\sqrt{2}}$$

$$\text{Now, } d \leq \sqrt{\frac{17}{2}} \Rightarrow \sqrt{(a-2)^2 + (7-a)^2} \cdot \frac{1}{\sqrt{2}} \leq \sqrt{\frac{17}{2}}$$

$$\Rightarrow a^2 - 9a + 18 \leq 0 \Rightarrow a \in [3, 6]$$



45. C

46. B

Sol. (for Q. 45.-46):

We can determine the value of $P_d(t, N)$ as = $\frac{\text{coefficient of } x^t \text{ in } (x^1 + x^2 + x^3 + \dots + x^d)^N}{d^N}$

$$\text{The value of } P_3(5, 2) = P_3(5, 3) = \frac{2}{9}; P_9(15, 2) = \frac{4}{81}$$

$$P_6(14, 3) = P_6(14, 5) = \frac{5}{72}; P_3(10, 4) = P(10, 6) = \frac{10}{81}$$

$$P_4(9, 3) = P_4(9, 4) = \frac{5}{32}$$

47. B

48. A

Sol. (for Q. 47.-48):

$$\text{Equation of line-1} \equiv \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$$

$$\text{Equation of line-2} \equiv 2x + 5y - z + 47 = 2x + y + z + 7 = 0 \text{ in symmetric form } \frac{x-\frac{3}{2}}{-3} = \frac{y+10}{2} = \frac{z}{4}$$

Shortest distance between both line

$$= \frac{|\vec{AB} \cdot (\vec{n}_1 \times \vec{n}_2)|}{|\vec{n}_1 \times \vec{n}_2|} = 3\sqrt{30}$$

$$\text{Now, P is point on AP} \Rightarrow P = (3 + 3r, 8 - r, 3 + r)$$

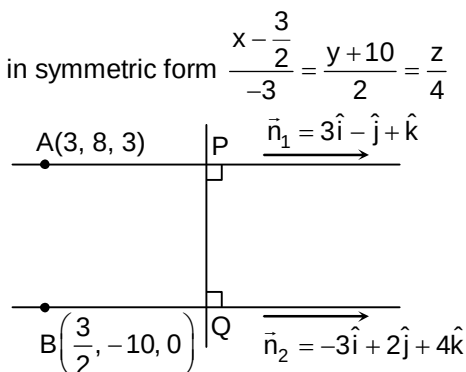
$$\text{Q is point of BQ} = Q = \left(\frac{3}{2} - 3r', -10 + 2r', 4r'\right)$$

$$\text{Now, } \vec{PQ} \parallel (\vec{n}_1 \times \vec{n}_2)$$

$$\Rightarrow \frac{\frac{3}{2} + 3r + 3r'}{-2} = \frac{19 - r - 2r'}{-5} = \frac{3 + r - 4r'}{1} \Rightarrow r = 0 \text{ and } r' = \frac{3}{2}$$

$$\therefore P = (3, 8, 3) \text{ and } Q = (-3, -7, 6) \text{ midpoint of PQ} = \left(0, \frac{1}{2}, \frac{9}{2}\right),$$

$$\text{so required plane } 2x + 5y - z + 2 = 0$$



Section – B

49. 3

 Sol. Let vertices $\vec{A}$, $\vec{B}$, $\vec{C}$ and circumcentre as origin, then

$$\vec{H} \text{ (orthocentre)} = \vec{A} + \vec{B} + \vec{C}$$

$$\vec{G} \text{ (centroid)} = \frac{\vec{H}}{3}$$

$$\vec{F} = \frac{\vec{G} + \vec{H}}{2} = \frac{2\vec{H}}{3}$$

$$\begin{aligned} \text{Now, } AF^2 + BF^2 + CF^2 &= (\vec{A} - \vec{F})(\vec{A} - \vec{F}) + (\vec{B} - \vec{F})(\vec{B} - \vec{F}) + (\vec{C} - \vec{F})(\vec{C} - \vec{F}) \\ &= \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} + \vec{C} \cdot \vec{C} - 2(\vec{A} + \vec{B} + \vec{C}) \cdot \vec{F} + 3\vec{F} \cdot \vec{F} = 3R^2 - \vec{F} \cdot (2(\vec{A} + \vec{B} + \vec{C}) - 3\vec{F}) = 3R^2 \end{aligned}$$

50. 2

Sol. Let point of intersection of line

$$y = \alpha x - \alpha + 2 \text{ with } y = x^2 \text{ are } x = x_1 \text{ and } x = x_2$$

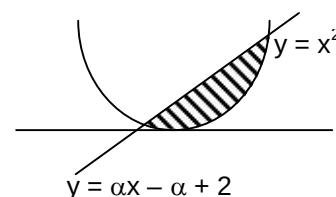
$$\text{Then, } x_1 + x_2 = \alpha ; x_1 x_2 = \alpha - 2$$

$$S(\alpha) = \int_{x_1}^{x_2} ((\alpha x - \alpha + 2) - x^2) dx$$

$$\text{For area to minimized } \frac{d(S(\alpha))}{d\alpha} = 0$$

$$\Rightarrow \frac{d(S(\alpha))}{d(\alpha)} = \frac{d}{d\alpha} \int_{x_1}^{x_2} ((\alpha x - \alpha + 2) - x^2) dx = \int_{x_1}^{x_2} \frac{d}{d\alpha} (\alpha x - \alpha + 2 - x^2) dx$$

$$= \int_{x_1}^{x_2} (x - 1) dx = (x_2 - x_1) \left(\frac{x_1 + x_2}{2} - 1 \right) = 0 \Rightarrow x_1 + x_2 = 2 \Rightarrow \alpha = 2$$



51. 1

$$\text{Sol. } m = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} ; n = \frac{1}{2} \sqrt{2a^2 + 2c^2 - b^2} ; p = \frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2}$$

$$\Rightarrow a^2 + b^2 + c^2 = (\lambda(2))^2 \{m^2 + n^2 + p^2\}$$

$$\Rightarrow a^2 + b^2 + c^2 = (\lambda(2))^2 \left\{ \frac{1}{4} \left\{ (2b^2 + 2c^2 - a^2) + (2a^2 + 2c^2 - b^2) + (2a^2 + 2b^2 - c^2) \right\} \right\}$$

$$\Rightarrow \lambda(2) = \frac{2}{\sqrt{3}}$$

Section – C

52. 02.00

53. 03.00

Sol. (for Q. 52.-53):

 Let the vertices of triangle ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3)

 Then x_1, x_2, x_3 are root of $a_1 x^3 + b_1 x + c_1 = 0$

$$\Rightarrow x_1 + x_2 + x_3 = 0 \text{ and } \sum x_1 x_2 = \frac{b_1}{a_1}.$$

Similarly $y_1 + y_2 + y_3 = 0$ and $\sum y_1 y_2 = \frac{b_2}{a_2}$

$\Rightarrow$ Circumcentre is at origin

$$\Rightarrow x_1^2 + y_1^2 = x_2^2 + y_2^2 = x_3^2 + y_3^2 = R^2 \Rightarrow x_1^2 + x_2^2 + x_3^2 + y_1^2 + y_2^2 + y_3^2 = 3R^2$$

$$\Rightarrow (x_1 + x_2 + x_3)^2 - 2\sum x_1 x_2 + (y_1 + y_2 + y_3)^2 - 2\sum y_1 y_2 = 3R^2$$

$$\Rightarrow -2\left[\sum x_1 x_2 + \sum y_1 y_2\right] = 3R^2 \Rightarrow -2\left[\frac{b_1}{a_1} + \frac{b_2}{a_2}\right] = 3R^2$$

54. 08.00

55. 04.00

Sol. Let point B(0, b) and C(0, c) and assume $b > c$

The equation of PB is $y - b = \frac{y_0 - b}{x_0}(x)$

$$\Rightarrow (y_0 - b)x - x_0 y + x_0 b = 0 \text{ as its tangent of circle } (x - 1)^2 + y^2 = 1$$

$$\Rightarrow 1 = \left| \frac{y_0 - b + x_0 b}{\sqrt{(y_0 - b)^2 + x_0^2}} \right| \Rightarrow (x_0 - 2)b^2 + 2y_0 b - x_0 = 0 \text{ similarly } (x_0 - 2)c^2 + 2y_0 c - x_0 = 0$$

$$\Rightarrow b + c = \frac{-2y_0}{x_0 - 2} \text{ and } bc = \frac{-x_0}{x_0 - 2} \Rightarrow (b - c)^2 = \frac{4x_0^2 + 4y_0^2 - 8x_0}{(x_0 - 2)^2} \Rightarrow b - c = \frac{2x_0}{x_0 - 2} \text{ (as } y_0^2 = 2x_0 \text{)}$$

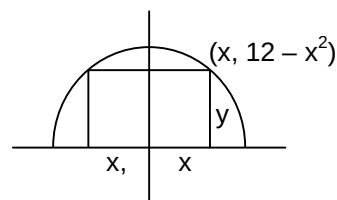
$$\text{Now, area of } \triangle PBC = \frac{1}{2}(b - c)x_0 = \frac{x_0^2}{x_0 - 2} = (x_0 - 2) + \frac{4}{(x_0 - 2)} + 4 \geq 8$$

56. 32.00

Sol. $A = 2x(12 - x^2)$

$$\frac{dA}{dx} = 2(12 - x^2 + (-2x)(2x)) = 24 - 6x^2$$

For maximum $x = 2$, $A = 32$ sq units



57. 86.50

(Range 86.50 to 86.60)

Sol. $A = 2x\sqrt{3}(10 - x)$

$$\frac{dA}{dx} = 2\sqrt{3}(10 - 2x)$$

Maximize when $x = 5$

$$A = 50\sqrt{3}$$

